



# Microbial carbon utilization in a boreal lake under the combined pressures of brownification and eutrophication: insights from a field experiment

Kevin Jones<sup>✉</sup> · Antonia Liess · Johanna Sjöstedt

Received: 30 May 2024 / Revised: 3 October 2024 / Accepted: 9 October 2024 / Published online: 21 October 2024  
© The Author(s) 2024

**Abstract** Climate and land use changes can increase terrestrial runoff to aquatic systems, leading to brownification and eutrophication in northern boreal lakes. Brownification may boost bacterial respiration and production, while eutrophication can enhance primary production and algal blooms. However, their combined effects on basal producers and bacterial carbon utilization are less understood. This study explores the combined impacts of the two stressors: brownification and eutrophication on microbial dynamics in Lake Bolmen. Utilizing a field mesocosm experimental design, treatments received different combinations of organic matter (OM) and inorganic nutrients to simulate predicted future scenarios. Results showed that OM additions significantly increased bacterial production and respiration, regardless of nutrient additions. Nutrient additions enhanced bacterial production but did not affect respiration. Both nutrients and OM stimulated bacterial

growth efficiency. Labile carbon from DOM was the main driver of higher bacterial respiration and short-term production increases. Fluorescence data indicated that the combination of brownification and eutrophication led to higher terrestrial DOM utilization than each stressor alone. The study suggests that future boreal lakes may become more heterotrophic, thus increasing CO<sub>2</sub> release. These findings highlight the complex interactions between DOM and nutrients and underscore the importance of considering multiple stressors in lake management and mitigation strategies.

**Keywords** Brownification · Eutrophication · Microbial dynamics · Carbon utilization · Organic matter

## Introduction

Climate change is a growing threat to the world's freshwater ecosystems (Woolway et al. 2020), especially at high latitudes. Many northern temperate boreal lakes are experiencing drastic changes in ecosystem function due to the loss of ice cover, changes in evaporation and water budgets, warming of surface waters, and alterations in mixing regimes (Woolway et al. 2020). In the boreal zone, climate change may further increase precipitation and thus organic matter (OM), nitrogen (N), and phosphorus (P) runoff from catchments into lakes (Kritzberg et al. 2014;

---

Handling Editor: Handling editor: Stefano Amalfitano.

**Supplementary Information** The online version contains supplementary material available at <https://doi.org/10.1007/s10750-024-05718-9>.

---

K. Jones (✉) · J. Sjöstedt  
Functional Ecology, Lund University, Lund, Sweden  
e-mail: kevin.jones@biol.lu.se

A. Liess · J. Sjöstedt  
School of Business, Innovation and Sustainability,  
Halmstad University, Halmstad, Sweden

Spilling et al. 2022). This increase in OM and nutrient loading may lead to darker lake water (brownification) and eutrophication (Tumledo 2010; Kritzberg 2017). Brownification is caused by refractory organic carbon (OC), such as humic compounds (Spilling et al. 2022), included in OM runoff and is affecting many boreal freshwater and coastal systems today (Temnerud et al. 2014; Taipale et al. 2016; Klante et al. 2021). In addition to climate change, brownification is further amplified by changes in land use in the catchment (afforestation) and the recovery from atmospheric sulfur deposition, which previously bound OM in soils (Mormul et al. 2012; Kritzberg 2017).

Brownification and eutrophication are two extensively studied stressors. However, their combined and interacting ecosystem consequences, together with the effects of climate change, are often difficult to predict. Brownification reduces light penetration and might thus shift pelagic basal production from autotrophy to heterotrophy (Deininger et al. 2016; Paczkowska et al. 2020; Figueroa et al. 2021), whereas eutrophication without browning may result in higher N input through runoff and atmospheric deposition, thus increasing lake water N:P ratios and instead boosting autotrophic phytoplankton production (Meunier et al. 2016). This increase in pelagic algal biomass increases light attenuation and hinders benthic autotrophic production (Taipale et al. 2018). Since eutrophication promotes pelagic autotrophic carbon fixation, it can also cause more labile autochthonous carbon release (Moss et al. 2011), which in turn may benefit bacterial production. Further, temperature increases due to climate change will prolong lake summer stratification, which hinders oxygenation of the deeper water layers and promotes bottom water anoxia. Anoxic bottom water causes sediments to release previously bound nutrients into the water column and potentially drives the system into a eutrophic state (Eriksson Wiklund et al. 2009; Brothers et al. 2014). Thus, to understand how lake ecosystems react to eutrophication and brownification, it is important to first understand how basal producers react to these two stressors separately and in combination.

Catchment soil OM is increasingly leaching into recipient aquatic systems, contributing to the dissolved OM (DOM) pool (Autio et al. 2016). The DOM pool is the largest bioavailable carbon pool in aquatic environments (Goodman et al. 2011)

and is classified according to its carbon content and origin. Autochthonous carbon is carbon produced by primary producers within the aquatic system, and allochthonous carbon is carbon produced outside the system (Kritzberg et al. 2004; Holding et al. 2017). Autochthonous carbon is the preferred energy source for bacteria due to its accessibility, whereas allochthonous carbon has a vast range of accessibilities to the microbial community (Spilling et al. 2022). Previously, allochthonous carbon was believed to be inaccessible to bacteria (recalcitrant) and thought to sediment in inland waters or be transported to coastal zones (Cole et al. 2007; Kritzberg et al. 2014). However, growing evidence suggests that not all terrestrial carbon is recalcitrant, and that remineralization may occur by light-mediated oxidation and bacterial breakdown (Cole 1999; Tranvik et al. 2009; Miranda et al. 2020).

Terrestrial DOM can be utilized and transformed into bacterial biomass or CO<sub>2</sub> through bacterial production (BP) and respiration (BR). Depending on which of these processes dominates, DOM metabolism can be a significant source of CO<sub>2</sub> to the atmosphere (Obernosterer and Benner 2004; Alleesson et al. 2021). Several studies have found a positive relationship between BP and DOM supply (Amon and Benner 1996; Eiler et al. 2003; Gonzalez et al. 2003), but a large number of studies have also failed to detect this positive relationship between BP and DOM concentration (Farjalla et al. 2002; Findlay 2003; Smith and Prairie 2004; Berggren et al. 2023). These inconsistent results suggest that factors besides bulk DOM content, such as DOM chemistry and nutrient availability, may be important in driving the metabolism of heterotrophic bacterioplankton (Berggren et al. 2023). In addition, bacterial respiration in lakes has been positively correlated with DOC (Hanson et al. 2003; Pace and Prairie 2005), nutrient concentration (Pace and Prairie 2005), and primary production (delGiorgio et al. 1996; Sand-Jensen and Staehr 2009), since bacteria may be limited by carbon or nutrients, or co-limited by both (Berggren et al. 2023). Additionally, nutrients and labile carbon may stimulate bacterial production of extracellular enzymes, leading to higher utilization of terrestrial and refractory carbon (Unanue et al. 1999; Arnosti 2011). This shows that brownification and eutrophication may interact (Kritzberg et al. 2020) and cause synergistic effects on carbon utilization.

Our study investigated bacterial carbon utilization in boreal lakes with respect to the predicted increase in eutrophication and brownification over the next 100 years. We hypothesized that (1) increased OM concentrations will stimulate bacterial production and respiration, and (2) treatments that received both OM and inorganic nutrients will utilize more of the terrestrial carbon pool compared to treatments with only OM addition. We studied this by subsampling a mesocosm experiment in Lake Bolmen, southern Sweden. The mesocosm experiment included treatments that represent eutrophication and brownification separately, as well as combined. OM was added in the form of a soil extract that was collected in Lake Bolmen's catchment. Nutrients were added as nitrate and phosphate. We tested our hypotheses by measuring BP, BR, bacterial growth efficiency (BGE), absorbance (watercolor), and fluorescence.

## Methods

### Description of study area and experimental facility

Lake Bolmen is southern Sweden's largest lake (184 km<sup>2</sup>) and stretches over three counties (Halland, Jönköping, and Kronoberg county). Its maximum depth is 37 m, and its retention time is 2.8 years. The Lake Bolmen catchment (1640 km<sup>2</sup>) consists of 48% forest, 22% marshland, 20% lakes, and 9% cultivated land (Tumdedo 2010). It is of special societal interest as a water reservoir for Scania. Lake Bolmen is a dimictic, oligotrophic brown water lake with ambient nutrient levels of total phosphorus (TP)  $\approx 15 \mu\text{g L}^{-1}$  and total nitrogen (TN)  $\approx 400 - 600 \mu\text{g L}^{-1}$ . Watercolor has increased from 20 – 40 mg Pt L<sup>-1</sup> in the 1980s to current levels of 80 mg Pt L<sup>-1</sup> in 2021. During the same period the TOC concentration are fluctuating between 9 and 11 mg L<sup>-1</sup> (Borgström 2020). Lake Bolmen is part of SITES (Swedish Infrastructure for Ecosystem Science) Aquanet, a standardized mesocosm infrastructure (Urrutia-Cordero et al. 2021) with one floating station placed in the southern part of Lake Bolmen, Halland county, Sweden (56.9428 N, 13.6456 E). The mesocosms, which are suspended in the lake utilizing the platform, are UV-resistant, translucent cylindrical polyethylene containers, each 1.5 m deep with a diameter of 0.8 m, and a total volume of 700 L (CPX Container 700 L

cylindrical, Cipax AB, Sweden). The mesocosms feature welded polyethylene extensions at both the top and bottom on all four opposing sides. At the bottom, a heavy iron chain serves as ballast to counteract buoyancy and enhance stability in the water. At the top, shackles are attached to the extensions, with straps threaded through and secured to the platform to keep the mesocosms in place. The mesocosms contained only water pumped in from the lake, with no sediment.

### Treatments

The experiment used 20 mesocosms, divided into five treatments with four replicates each: (1) control, (2) OM low N:P ratio (OM) treatment, (3) OM high N:P ratio (OM+HNP) treatment, (4) low N:P ratio (LNP) treatments, and (5) high N:P ratio (HNP) treatment (Figure. S1). The control contained only Lake Bolmen water. The OM and OM+HNP treatments contained Lake Bolmen water with added soil extract to attain a watercolor of  $\approx 180 \text{ mg L}^{-1}$  Pt (predicted for Lake Bolmen by the year 2120). The LNP treatments received dissolved inorganic P as well as dissolved inorganic N to obtain similar concentrations of TN and TP as the OM treatment. The HNP treatment had a higher N:P ratio and to achieve this more N was added together with the same P addition as in the LNP treatment (Table. S2). In the OM+HNP treatment, inorganic nutrient concentration was adjusted to similar levels as the HNP treatment (Table. S2). The purpose of this was to create comparable situations between treatments OM vs LNP and OM+HNP vs HNP, since the addition of slurry brought along organic and inorganic nutrients. The differing N:P ratios were used as well to allow for the possibility of investigating current concentrations versus projected future concentrations.

### Soil extract and implementation of the treatments

Soil was collected within the Lake Bolmen catchment, roughly representing the different proportions of land use types. After soil collection, a soil extract was produced according to standard methods (Liess et al. 2015). First, large objects were removed, then the soil was homogenized and pushed through a 1 mm size mesh. The remaining homogenized soil was dried for 48 h at room temperature. Then 2 kg of

soil, 13.33 l of distilled water, and 467 g of chelating ion exchange media (Supelco Diaion CR11 sodium form) were mixed and left in the dark for 48 h. Afterwards, the slurry was sieved through a 100 µm soft fabric mesh.

The resulting soil extract (4.5 l) was added to each of the mesocosms in the treatments OM and OM+HNP. Nitrogen was amended in the form of dissolved  $\text{NO}_3\text{-N}$  ( $\text{NaNO}_3$ ) and added to the LNP (40 mg  $\text{NO}_3\text{-N}$ ), HNP (252 mg  $\text{NO}_3\text{-N}$ ), and the OM+HNP (210 mg  $\text{NO}_3\text{-N}$ ) treatments. Phosphorus was added as dissolved phosphate  $\text{PO}_4\text{-P}$  in the form of ( $\text{Na}_2\text{HPO}_4$ ) to treatments LNP (53 mg  $\text{PO}_4\text{-P}$ ) and HNP (53 mg  $\text{PO}_4\text{-P}$ ). The six-week mesocosm experiment was conducted from July 6th to August 16th in 2021. On July 5th (day -1), the mesocosms were filled with 500 l of lake water. The next day (day 0), treatments were applied. Surface water temperature was measured ( $\approx 20^\circ\text{C}$ ), then water was collected from the mesocosm experiment on day 1 and transported to the laboratory at Lund University (Department of Biology, Aquatic Ecology) where incubations and analysis were conducted. Once at the laboratory water from all mesocosms was filtered through A/E glass fiber filters (142 mm, 1 µm pore size, PALL Corporation). The purpose was to remove any large particles and to act as a pre-filtration to prevent filter clogging at later finer filtering steps. Filtrates were collected in acid-washed 5 l polyurethane bottles.

Bacterial respiration (BR), bacterial production (BP) and bacterial growth efficiency (BGE)

Water was subsampled from the respective 5 l polyurethane bottles and bacterial oxygen demand was examined through the change in dissolved oxygen (DO) over a 5-day incubation period at  $20^\circ\text{C}$  in the dark (Sawyer et al. 1978; Wilhelm 2009). Water from each mesocosm was carefully added into 2 L glass bottles to avoid bubbles and headspace and finally capped with glass stoppers, in addition, 3 controls with milli-Q water were incubated. Oxygen ( $\text{mg L}^{-1}$ ) was measured every 30 min using optical oxygen sensors (optodes) and three 10-channel fiber oxygen transmitters (oxy-10, PreSens®). The first 14.5 h of stabilization, as the bottles' temperature adjusted, were excluded from the data analysis resulting in a 97.5 h incubation time. Oxygen concentration ( $\text{mg L}^{-1}$ ) was plotted over time, starting from 8.5 to

$6.5 \text{ mg L}^{-1}$  and ending between 7.6 and  $4.6 \text{ mg L}^{-1}$ . The linear slope was calculated to determine the oxygen consumption rate ( $\text{mg O}_2 \text{ L}^{-1} \text{ Day}^{-1}$ ). The oxygen consumption rate was converted to carbon units using a molar respiratory quotient of 1.0 to obtain the BR (Soares et al. 2019). The bioavailable fraction was based on BR data and calculated by dividing the amount of carbon utilized for respiration (converted oxygen consumption rate) within a day by the TOC concentration. BP was measured at the start and end of the incubation period, following the protocol by Smith and Azam (Smith and Azam 1992). All samples were processed in technical triplicates. Labeled leucine (L- [4, 5- $^3\text{H}$ ], Perkin Elmer) was diluted with unlabeled leucine (Sigma, St. Louis, MO, USA) and added at a final concentration of 100 nM. The samples were washed once with 5% trichloroacetic acid and twice with ethanol before adding the scintillation cocktail. Disintegrations per minute were recalculated to bacterial carbon production rates ( $\mu\text{mol C L}^{-1} \text{ h}^{-1}$ ) (Kirchman 1993). Bacterial growth efficiency (BGE) was calculated based on the BP and BR measurements ( $\text{BGE} = \text{BP} / [\text{BP} + \text{BR}]$ ) (Muscarella et al. 2020).

#### Nutrient and TOC analysis

Total nitrogen (TN) and TOC samples were subsampled into acid-washed 50 ml Falcon tubes at the start and end of the incubation for BR. Total phosphorus (TP) samples were collected in acid-washed 50 ml Falcon tubes at the start of the incubation. TN and TOC samples were acidified and analyzed at the Department of Biology, Lund University, using a TOC V-CPH/CPN (Shimadzu, Tokyo, Japan) via the non-purgeable organic carbon method. TP was analyzed using the molybdate reactive phosphorus method (Menzel and Corwin 1965).

#### Absorbance and fluorescence

Subsamples were obtained from each incubation bottle at the start and the end of the incubation period. Samples used for water color were filtered through a  $0.45 \mu\text{m}$  filter ( $0.45 \mu\text{m}$  Supor Acrodisc, Cytiva) and analyzed on a UV-2600, UV-VIS spectrophotometer (Shimadzu, Tokyo, Japan) at 420 nm using a 5 cm quartz cuvette and milli-Q water as a blank (Pennanen 1986) (Table 1).

**Table 1** DOM proxies and corresponding absorbance/excitation and emissions

| DOM Composition | Source/Proxy                   | Absorbance/<br>Excitation & Emission |
|-----------------|--------------------------------|--------------------------------------|
| a420            | Watercolor                     | A420*500                             |
| Peak A (R.U.)   | Humic-like: terrestrial origin | Ex: 260<br>Em: 400–460               |
| Peak C (R.U.)   | Humic-like: terrestrial origin | Ex: 320–360<br>Em: 420–460           |
| Peak B (R.U.)   | Tyrosine/protein-like          | Ex: 275<br>Em: 305                   |
| Peak T (R.U.)   | Tryptophan/protein-like        | Ex: 275<br>Em: 340                   |

\*R.U. = Raman Units

Fluorescence samples were collected from each incubation bottle at the start and end of the incubation period. Collected samples were filtered through a 0.2  $\mu\text{m}$  filter (0.2  $\mu\text{m}$  Supor Acrodisc, Pall Corporation) and collected in 50 ml acid-washed and combusted brown glass vials. Samples were analyzed using a Horiba Aqualog fluorometer with a 1 cm quartz cuvette at Chalmers University in Gothenburg, Sweden. The method used for the Aqualog was a 3D measurement (Fluorescence excitation-emission matrices [EEM] + Absorbance), Ex: 600 to 240 in 5 nm increments, Em: 600–240 nm, 8 px binning (4 nm increments), medium gain, and an integration time of 1 s. Samples were processed using the drEEM v0.6.5 toolbox in MATLAB v9.12 (MathWorks Inc) (Murphy et al. 2013). EEMs were processed by blank EEM subtraction, correcting inner filter effects, Raman and Rayleigh scatter, and EEMs were normalized to Raman Units (R.U.) (Wünsch and Murphy 2021). Corrected EEMs were analyzed with parallel factor analysis (PARAFAC), and commonly reported fluorescence peaks (A, C, B, T) were investigated to further understand changes in OM composition (Coble 1996) (Table 1). To investigate the change in OM composition, fluorescence EEM data were grouped into treatments and by timepoint (start and end). End timepoint EEMs were subtracted by the start timepoint EEMs, resulting in the change of fluorescent intensity from start to end for each treatment (Fig. S2).

### Statistics

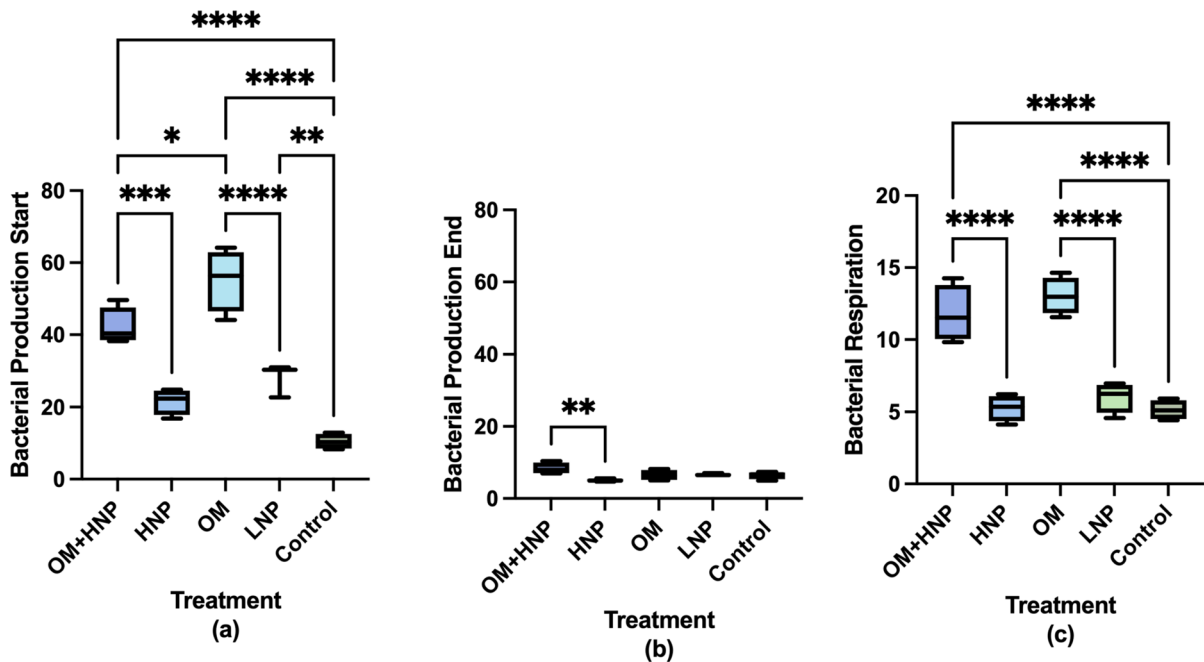
All statistical analyses were performed using Prism GraphPad version 9.2.0 and MATLAB 9.12.0.2170939 (R2022a) Update 6. In Prism,

one-way ANOVAs assessed differences between the five treatments for BP, BR, and BGE, as well as changes in watercolor before and after incubation within treatments. In MATLAB, paired sample t-tests checked for significant changes in fluorescence intensity of OM composition (peaks A, C, B, T) before and after incubation within treatments.

### Results

Bacterial production (BP), bacterial respiration (BR) and bacterial growth efficiency (BGE)

OM addition significantly increased BP, BR and BGE. At the start of the incubations, BP was significantly higher in OM+HNP ( $38\text{--}50\ \mu\text{mol C L}^{-1}\text{ Day}^{-1}$ ) and OM treatments ( $44\text{--}64\ \mu\text{mol C L}^{-1}\text{ Day}^{-1}$ ) than in the treatments without OM addition (control:  $8\text{--}13\ \mu\text{mol C L}^{-1}\text{ Day}^{-1}$ ; LNP:  $22\text{--}31\ \mu\text{mol C L}^{-1}\text{ Day}^{-1}$  and HNP:  $16\text{--}25\ \mu\text{mol C L}^{-1}\text{ Day}^{-1}$ , Fig. 1a, ANOVA,  $p < 0.001$ ). BP in the LNP treatment was significantly higher than in the control treatment (Fig. 1a, ANOVA,  $p = 0.005$ ). However, there was no significant difference in BP between the HNP treatment and the control (Fig. 1a, ANOVA,  $p = 0.066$ ). In all but the control, BP significantly decreased from start to end of the incubations (ANOVA,  $p < 0.0001$ ). After the incubation, only a significant difference between the OM+HNP ( $7\text{--}10\ \mu\text{mol C L}^{-1}\text{ Day}^{-1}$ ) and HNP ( $5\text{--}6\ \mu\text{mol C L}^{-1}\text{ Day}^{-1}$ ) treatment remained (Fig. 1b, ANOVA,  $p = 0.005$ ). BR was significantly higher in the OM+HNP ( $9.8\text{--}14.3\ \mu\text{mol C L}^{-1}\text{ Day}^{-1}$ ) and OM ( $11.5\text{--}14.6\ \mu\text{mol C L}^{-1}\text{ Day}^{-1}$ ) treatments than in the control, HNP and LNP ( $\approx 4\text{--}7\ \mu\text{mol C L}^{-1}\text{ Day}^{-1}$ ) treatments (Fig. 1c, ANOVA,



**Fig. 1** Bacterial production (BP) of the different treatments at the start of the incubation (a), at the end of the incubation (b) and bacterial respiration (BR, based on oxygen consumption

between start and end of incubation) (c). (ANOVA, \*  $p \leq 0.05$ , \*\*  $p \leq 0.01$ , \*\*\*  $p \leq 0.001$ , \*\*\*\*  $p \leq 0.0001$ ). Units for y-axis are,  $\mu\text{mol C L}^{-1} 24 \text{ h}^{-1}$

$p < 0.0001$ ). There were no significant differences in BR between the treatments with only inorganic nutrient additions (HNP and LNP) and the control treatments (Fig. 1c). BGE was significantly enhanced by both inorganic nutrient and OM addition at the start of the incubation (Fig. 2a, ANOVA,  $p \leq 0.01$ ). In the control treatment, BGE ranged from 0.6 to 0.71 and in the other treatments from 0.76 to 0.91 (Fig. 2a). BGE decreased after incubation in all treatments (Fig. 2b) and was significantly lower in treatments with OM additions (OM+HNP and OM) than in the control, LNP and HNP treatments (Fig. 2b, ANOVA,  $p \leq 0.05$ ). In addition, BGE was significantly higher in the OM+HNP than in the OM treatment (Fig. 2b, ANOVA,  $p \leq 0.05$ ).

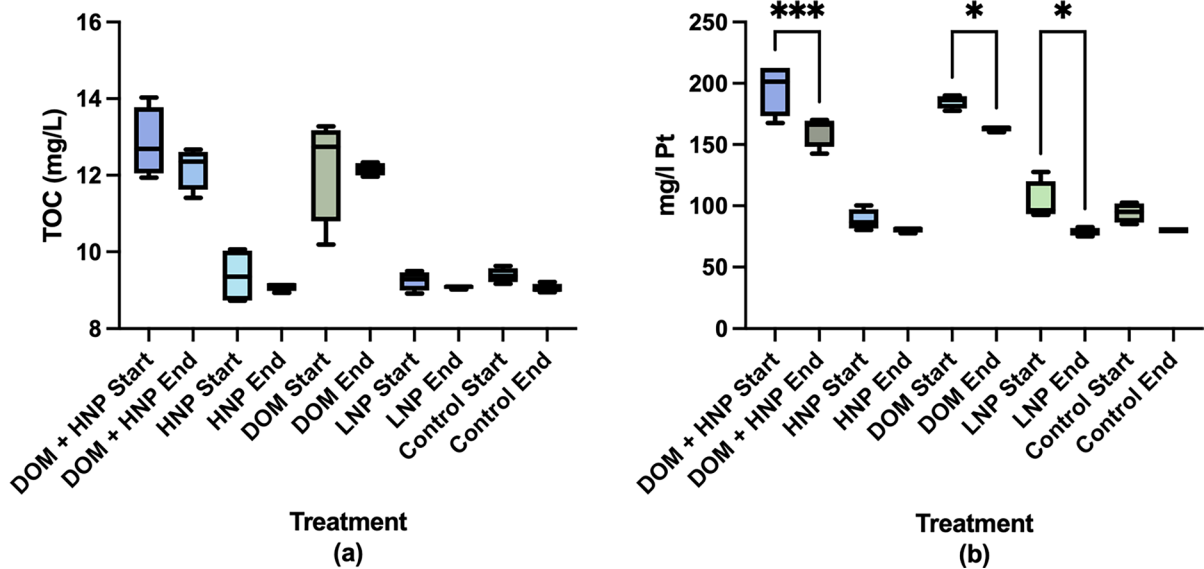
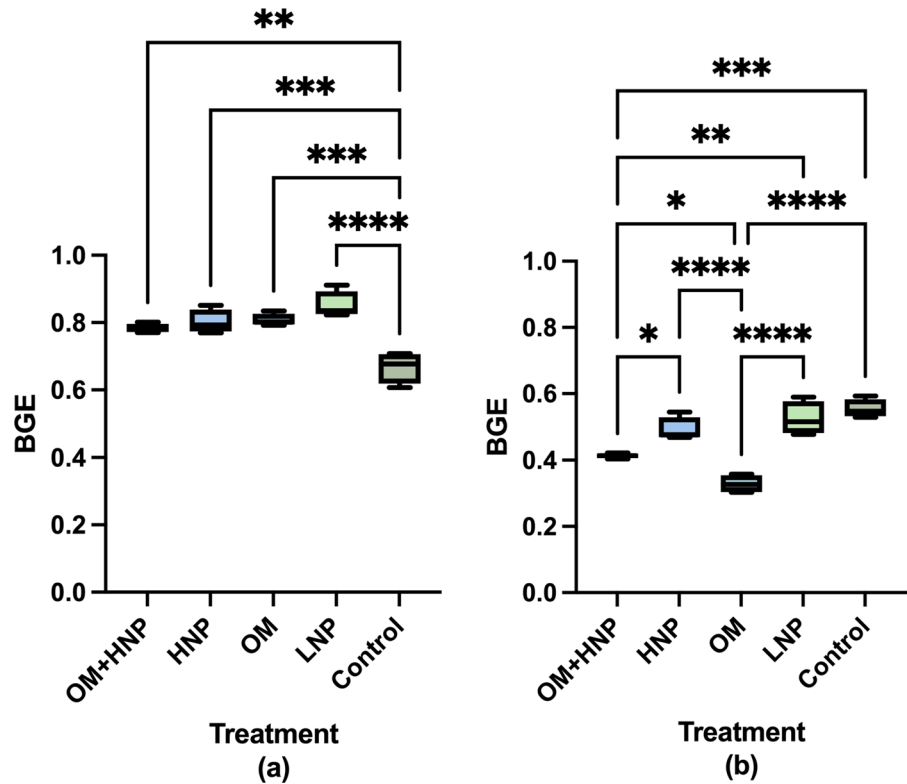
#### TOC utilization

Average TOC concentrations decreased from start and end of the incubations. Over the 5-day incubation, bacteria in the OM+HNP, HNP, OM, LNP, and control treatments consumed 5%, 3%, 0.6%, 1.8% and 3.6% of the bulk TOC pool, respectively (Table. S1). However, due to large variations in

initial values in each treatment, the change in TOC concentration between start and end of the incubation was not statistically significant (Fig. 3a). In contrast, the addition of OM significantly increased the bioavailable fraction used for respiration in the OM+HNP (0.9 – 1.2%) and OM (1 – 1.6%) treatments compared to the control (0.6 – 0.74%), HNP (0.6 – 0.74%) and LNP (0.6 – 0.9%) treatments (Fig. S3 ANOVA,  $p \leq 0.05$ ).

Watercolor was examined before and after the 5-day incubation to examine the change in concentration of the colored dissolved organic matter (cDOM) and the aromatic content of the DOM. Watercolor was significantly higher in the treatments with OM additions (OM and OM+HNP) than in the control, LNP and HNP treatments, both before and after the incubation (Fig. 3b) (ANOVA,  $p \leq 0.0001$ ). The OM+HNP, OM and LNP treatments yielded significant (ANOVA,  $p \leq 0.05$ ) reductions in watercolor averaging a decrease of  $34 \text{ mg Pt L}^{-1}$ ,  $21 \text{ mg Pt L}^{-1}$  and  $24 \text{ mg Pt L}^{-1}$ , respectively (Fig. 3b). There were no significant differences in watercolor between start and end of incubation for the HNP treatment and the control (Fig. 3b).

**Fig. 2** Bacterial growth efficiency (BGE) of the different treatments at the start of the incubation (a) and at the end of the incubation (b). \* Denote significant differences (ANOVA, \*  $p \leq 0.05$ , \*\*  $p \leq 0.01$ , \*\*\*  $p \leq 0.001$ , \*\*\*\*  $p \leq 0.0001$ )



**Fig. 3** Change in TOC concentration (a) and watercolor (b) for each treatment at the start and end of incubation. Statistics were conducted within treatments to examine changes between

start and end. Only significant differences between start and end values (not between treatments) are shown in the graph. (ANOVA, \*  $p \leq 0.05$ , \*\*  $p \leq 0.01$ , \*\*\*  $p \leq 0.001$ )

## Fluorescence excitation–emission matrix-parallel factor analysis (EEM-PARAFAC)

The PARAFAC model resulted in a split validated 4 component model (Fig. S4) which corresponded to common peak regions (PeakA, PeakC, PeakB, and Peak T) (Coble 1996). Fluorescence properties of DOM were altered during incubation and decreases were observed for all peaks in all treatments, except for peakA in the OM treatment where a production of this peak occurred (Table 2). Controls showed a significant decrease in fluorescent intensity from start to end of incubation for peakA ( $t$ -test,  $p=0.046$ ), peakC ( $t$ -test,  $p=0.001$ ), and peakT (Table 2,  $t$ -test,  $p=0.010$ ). LNP treatments had a significant decrease between start and end of incubation in peakC (Table 2,  $t$ -test,  $p=0.0127$ ) and peakT (Table 2  $t$ -test,  $p=0.034$ ). OM+HNP treatments had significant decreases for peakA (Table 2,  $t$ -test,  $p=0.046$ ), peakC (Table 2,  $t$ -test, 0.001), and peakT (Table 2,  $t$ -test,  $p=0.014$ ) whereas OM treatments had a significant decrease in peakT (Table 2,  $t$ -test,  $p=0.045$ ). There were no significant changes in fluorescent intensity for peakB in any of the treatments (Table 2).

## Discussion

Our results show that OM additions significantly enhanced BP and BR regardless of inorganic nutrient additions, supporting hypothesis 1. Furthermore, the treatment receiving both OM and inorganic nutrients (OM+HNP) utilized more of the terrestrial carbon pool than the treatment with only OM addition, supporting hypothesis 2. OM additions provided both labile carbon and accessible nutrients, stimulating

bacterial production and respiration, with extra added nutrients promoting terrestrial TOC utilization. Our study further shows that nutrient addition stimulated BP but not BR, potentially indicating that nutrient addition shifted bacteria from being nutrient-limited to carbon-limited (Cimblaris and Kalff 1998). OM addition significantly increased BP ( $\approx 95\%$ ) and BR ( $\approx 63\%$ ) compared to corresponding nutrient treatments. This was likely due to the bacterial community in Lake Bolmen being co-limited by carbon and inorganic nutrients (Zweifel et al. 1993; Dorado-García et al. 2014; Berggren et al. 2023), due to the carbon pool in the lake being terrestrially derived and recalcitrant.

Bacterial growth efficiency (BGE) reflects the relationship between BP and BR (Muscarella et al. 2020). At the start of the incubation, BGE was on average 50% higher in all treatments compared to the control treatment. This indicates that both inorganic nutrient and OM addition stimulated BP more than BR. However, this temporary boost in BGE diminished by the end of the incubation, with BGE significantly lower in the treatments with OM additions. The high BGE at the beginning of the incubation (approximately 80%) is likely an overestimation due to our sample collection method. It is possible that we missed the initial peak in BR but captured the peak in BP, resulting in an overestimation of BGE. A previous study found a more delayed response for BP than for BR after OM addition. Bacteria first used the labile fraction of the added carbon for BR to obtain the required energy to produce ectoenzymes, allowing them to further break down refractory carbon, resulting in an increase in biomass (Pérez and Sommaruga 2006). In our study, the pulse of OM significantly increased bacterial

**Table 2** Fluorescent change in common peaks (A, C, B, and T) from start to end of incubation (5 days). MeanPeak (A, C, B, and T) is the mean value of change in fluorescence intensity of the 4 samples per treatment. Peakstd (A, C, B, and T) is

the standard deviation between the subtracted EEMS, and Pv (A, C, B, T) are the results from a  $t$ -test examining differences between start and end in these common peaks per treatment. (\* indicates a significant decrease from start to end of incubation)

| Treatment | Mean<br>PeakA | PeakA<br>std | PvA    | Mean<br>PeakC | PeakC<br>std | PvC    | Mean<br>PeakB | PeakB<br>std | PvB   | Mean<br>PeakT | PeakT<br>std | PvT    |
|-----------|---------------|--------------|--------|---------------|--------------|--------|---------------|--------------|-------|---------------|--------------|--------|
| OM+HNP    | −0.090        | 0.055        | 0.046* | −0.035        | 0.006        | 0.001* | −0.037        | 0.050        | 0.232 | −0.046        | 0.018        | 0.014* |
| HNP       | −0.080        | 0.076        | 0.124  | −0.048        | 0.038        | 0.089  | −0.044        | 0.046        | 0.148 | −0.054        | 0.041        | 0.079  |
| OM        | 0.014         | 0.045        | 0.571  | −0.013        | 0.013        | 0.153  | −0.049        | 0.063        | 0.220 | −0.070        | 0.042        | 0.045* |
| LNP       | −0.078        | 0.061        | 0.083  | −0.033        | 0.016        | 0.013* | −0.013        | 0.027        | 0.401 | −0.045        | 0.024        | 0.034* |
| Control   | −0.061        | 0.037        | 0.046* | −0.016        | 0.010        | 0.047* | −0.026        | 0.023        | 0.116 | −0.024        | 0.008        | 0.010* |

activity (BP and BR), but after 5 days, BP returned to levels corresponding to the control treatment. This indicates that the bacterial community of Lake Bolmen can degrade fractions of terrestrial matter very quickly. Additionally, previous studies have shown that OM and nutrient additions affect BP and BR rates differently (Lennon and Pfaff 2005; Soares et al. 2018), which might explain why BGE can be high in natural settings during short periods while bacterial communities utilize labile fractions of carbon and nutrients (Eiler et al. 2003; Jansson et al. 2006).

In our study, bacteria consumed a much lower fraction of the bulk TOC than reported in previous studies (Sondergaard and Middelboe 1995; Lennon and Pfaff 2005), likely due to differences in the duration of incubation (Berggren et al. 2023) or OM composition (Kothawala et al. 2014; Autio et al. 2016; Jones et al. 2023; Zhao et al. 2023). Although OM additions have previously been shown to stimulate OM consumption (Traving et al. 2017), examining the TOC consumption for all treatments, the OM+HNP (brownification+eutrophication) treatment was the only one where bacteria consumed more of the total TOC pool than in the control. Absorbance and fluorescence data revealed that terrestrial OM was utilized in all treatments, but the utilization of refractory OM was higher in the OM+HNP treatment than in all other treatments. When examining the changes in fluorescent intensities of both terrestrial peaks (A and C), only the control and OM+HNP treatments significantly reduced both. There are two reasons for bacteria to invest energy in utilizing terrestrial TOC: either they need nutrients, or they need carbon (Guenet et al. 2010). One possibility is that TOC in the control is utilized to release nutrients, whereas in the OM+HNP treatment, TOC is utilized to obtain carbon. Comparing the OM utilization in the treatments with OM additions, the OM+HNP treatment has both a higher percentage of TOC utilized and a significant reduction in the intensity of fluorescence peaks A and C. In contrast, for the OM treatment, only peak T (protein-like compounds) was significantly reduced. This confirms that a facilitative effect is occurring with the addition of high nutrient concentrations and OM (Dorado-García et al. 2014; Thompson and Cotner 2020). We believe our data demonstrate the importance of nutrient availability for terrestrial TOC utilization (Fonvielle et al. 2016).

The addition of OM had a greater impact than the addition of only inorganic nutrients on bacterial processes (Fig. 1, OM additions vs LNP and HNP). This contradicts the results from Zhao et al. (2023) who compared the impact of eutrophication and increased input of refractory terrestrial compounds in the coastal northern Baltic Sea. Here, plankton extract was used to simulate eutrophication, as it causes increased primary production, augmenting labile autochthonous carbon availability. This plankton extract addition induced much larger effects on the bacterial community than allochthonous terrestrial DOM addition (Zhao et al. 2023). Differences between our study and (Zhao et al. 2023) may be due to differences in experimental design, with eutrophication in our experiment being simulated by the addition of inorganic nutrients. While our study focused closely on the immediate effects of brownification and eutrophication, nutrient uptake by phytoplankton may lead to release of autochthonous carbon over time and thus fuel bacterioplankton production. Phosphorus addition has been shown to accelerate the release of autochthonous carbon in a mesocosm study investigating DOM dynamics, but this effect was seen first after 3 to 6 weeks (Fonvielle et al. 2016). The opposite effect, with bacterioplankton stimulating phytoplankton succession, was observed in a mesocosm experiment by Traving et al. (2017), where additions of DOM increased microbial activity and caused a release of inorganic nutrients. We conclude that the effects of brownification and eutrophication are linked and that their combined impact will have consequences for nutrient biogeochemistry and basal producers, potentially causing bottom-up effects in the lake food web.

Brownification has been an ongoing issue in Lake Bolmen since the 1980s (Klante et al. 2021), and the bacterial communities are likely already adapted to metabolizing complex organic molecules. This may result in only subtle changes in community composition or function as brownification progresses. These communities likely possess enzymes and metabolic pathways optimized for processing allochthonous organic matter, leading to greater stability and resilience to additional OM inputs (Traving et al. 2017). In contrast, bacterial communities in clear lakes, which typically rely more on autochthonous organic matter, could undergo significant shifts in composition as brownification alters the availability of resources

and environmental conditions (Kritzborg et al. 2014). The introduction of high levels of OM reduces light penetration and oxygen levels, particularly in deeper waters, forcing a transition from aerobic to anaerobic microbial processes (Kritzborg et al. 2014). This could have broader impacts on the lake's ecosystem, including changes in nutrient cycling, primary production, and food web dynamics. Additionally, while the bacterial communities in brown lakes may exhibit resilience due to long-term exposure to these conditions, those in clear lakes might experience more pronounced and rapid shifts, potentially leading to altered ecosystem functions (Carpenter et al. 2011). While mesocosm experiments are generally effective at representing natural conditions and providing the control needed for experiments, they have limitations and may not capture all the complexities of real-world ecosystems. Additionally, the experimental design can significantly influence the results. Filtration will, for example remove part of the bacterial community and larger predators which could influence the bacterial processes. In addition, this study only focused on the effect on microbial degradation of carbon. In natural environment photooxidation is also an important process for the transformation of organic matter leading to an increased production of labile carbon, further fueling bacterial production resembling the effects of eutrophication resulting in an increase in autochthonous carbon production. This interplay between brownification and eutrophication may result in a feedback loop that enhances bacterial respiration and CO<sub>2</sub> release, contributing to further climate warming. In mesotrophic boreal lakes, the transition to eutrophic conditions driven by brownification and eutrophication may lead to profound ecological impacts, including altered nutrient cycling, increased algal blooms, and potential hypoxia or anoxia in deeper waters, further exacerbating eutrophication and its cascading effects on ecosystem dynamics (Kritzborg et al. 2020).

## Conclusion

Climate change in northern Europe may lead to warming and increased precipitation, which can enhance both eutrophication and brownification. We found that the immediate effect of brownification on BP and BR was stronger than the effect of eutrophication. In

addition, our study provides support for coalescing effects of eutrophication and brownification leading to higher carbon utilization in boreal brown water lakes. The alteration of bacterial dynamics could proliferate anoxia in bottom waters, affect lake ecosystem services, change the lake food web, and shift biogeochemical cycles. Our study provides evidence that increased eutrophication will facilitate carbon utilization, leading to a greater release of CO<sub>2</sub> into the atmosphere, thus further driving climate change.

**Acknowledgements** We would like to thank: Deborah Dupont, Tatyana Barnes, Anna Borgström and Juha Rankinen for their help and hard work setting up the mesocosm experiment. SITES aquanet for allowing us to use their infrastructure. Kathleen Murphy for lending us their Aqualog, and Martin Berggren for the use of their Oxy-10 sensors.

**Author Contributions** All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Kevin Jones and Johanna Sjöstedt. The first draft of the manuscript was written by Kevin Jones, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

**Funding** Open access funding provided by Lund University. This research was supported by the Swedish Research Council FORMAS (2020–00730) and Carl Tryggers foundation (CTS 19:337) to Johanna Sjöstedt.

**Data Availability** The datasets generated during and/or analyzed during the current study is publicly available on Mendeley data, DOI: <https://doi.org/10.17632/gmv4m94wvc.1>, also available from the corresponding author on reasonable request.

## Declarations

**Conflict of interests** The authors have no relevant financial or non-financial interests to disclose.

**Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

## References

- Allesson, L., B. Koehler, J. E. Thrane, T. Andersen & D. O. Hessen, 2021. The role of photomineralization for CO<sub>2</sub> emissions in boreal lakes along a gradient of dissolved organic matter. *Limnology and Oceanography* 66(1): 158–170. <https://doi.org/10.1002/lno.11594>.
- Amon, R. & R. Benner, 1996. Bacterial utilization of different size classes of dissolved organic matter. *Limnology and Oceanography* 41(1): 41–51.
- Arnosti, C., 2011. Microbial extracellular enzymes and the marine carbon cycle. *Annual Review of Marine Science* 3(1): 401–425. <https://doi.org/10.1146/annurev-marine-120709-142731>.
- Autio, I., H. Soinne, J. Helin, E. Asmala & L. Hoikkala, 2016. Effect of catchment land use and soil type on the concentration, quality, and bacterial degradation of riverine dissolved organic matter. *Ambio* 45(3): 331–349. <https://doi.org/10.1007/s13280-015-0724-y>.
- Berggren, M., L. Ye, R. A. Sponseller, A.-K. Bergström, J. Karlsson, H. Verheijen & G. Hensgens, 2023. Nutrient limitation masks the dissolved organic matter composition effects on bacterial metabolism in unproductive freshwaters. *Limnology and Oceanography* 68(9): 2059–2069. <https://doi.org/10.1002/lno.12406>.
- Borgström, A., 2020. Lake Bolmen: Past, present and future. Lund University, 32.
- Brothers, S., J. Köhler, K. Attemeyer, H. P. Grossart, T. Mehner, N. Meyer, K. Scharnweber & S. Hilt, 2014. A feedback loop links brownification and anoxia in a temperate, shallow lake. *Limnology and Oceanography* 59(4): 1388–1398. <https://doi.org/10.4319/lno.2014.59.4.1388>.
- Carpenter, S. R., E. H. Stanley & M. J. Vander Zanden, 2011. State of the world's freshwater ecosystems: physical, chemical, and biological changes. *Annual Review of Environment and Resources* 36(1): 75–99. <https://doi.org/10.1146/annurev-environ-021810-094524>.
- Cimbleiris, A. C. P. & J. Kalff, 1998. Planktonic bacterial respiration as a function of C:N:P ratios across temperate lakes. *Hydrobiologia* 384(1): 89–100. <https://doi.org/10.1023/A:1003496815969>.
- Coble, P. G., 1996. Characterization of marine and terrestrial DOM in seawater using excitation-emission matrix spectroscopy. *Marine Chemistry* 51(4): 325–346. [https://doi.org/10.1016/0304-4203\(95\)00062-3](https://doi.org/10.1016/0304-4203(95)00062-3).
- Cole, J. J., 1999. Aquatic microbiology for ecosystem scientists: new and recycled paradigms in ecological microbiology. *Ecosystems* 2(3): 215–225. <https://doi.org/10.1007/s100219900069>.
- Cole, J. J., Y. T. Prairie, N. F. Caraco, W. H. McDowell, L. J. Tranvik, R. G. Striegl, C. M. Duarte, P. Kortelainen, J. A. Downing, J. J. Middelburg & J. Melack, 2007. Plumbing the global carbon cycle: Integrating inland waters into the terrestrial carbon budget. *Ecosystems* 10(1): 171–184. <https://doi.org/10.1007/s10021-006-9013-8>.
- Deininger, A., C. L. Faithfull, K. Lange, T. Bayer, F. Vidussi & A. Liess, 2016. Simulated terrestrial runoff triggered a phytoplankton succession and changed seston stoichiometry in coastal lagoon mesocosms. *Marine Environmental Research* 119: 40–50. <https://doi.org/10.1016/j.marenvres.2016.05.001>.
- delGiorgio, P., D. F. Bird, Y. T. Prairie & D. Planas, 1996. Flow cytometric determination of bacterial abundance in lake plankton with the green nucleic acid stain SYTO 13. *Limnology and Oceanography* 41(4): 783–789.
- Dorado-García, I., J. M. Medina-Sánchez, G. Herrera, M. J. Cabrerizo & P. Carrillo, 2014. Quantification of carbon and phosphorus co-limitation in bacterioplankton: new insights on an old topic. *PLOS ONE* 9(6): e99288. <https://doi.org/10.1371/journal.pone.0099288>.
- Eiler, A., S. Langenheder, S. Bertilsson & L. J. Tranvik, 2003. Heterotrophic bacterial growth efficiency and community structure at different natural organic carbon concentrations. *Applied and Environmental Microbiology* 69(7): 3701–3709. <https://doi.org/10.1128/aem.69.7.3701-3709.2003>.
- Eriksson Wiklund, A., K. Dahlgren, B. Sundelin & A. Andersson, 2009. Effects of warming and shifts of pelagic food web structure on benthic productivity in a coastal marine system. *Marine Ecology Progress Series* 396: 13–25. <https://doi.org/10.3354/meps08290>.
- Farjalla, V., B. Faria & F. Esteves, 2002. The relationship between DOC and planktonic bacteria in tropical coastal lagoons. *Archiv Für Hydrobiologie* 156: 97–119. <https://doi.org/10.1127/0003-9136/2002/0156-0097>.
- Figueroa, D., E. Capo, M. V. Lindh, O. F. Rowe, J. Paczkowska, J. Pinhassi & A. Andersson, 2021. Terrestrial dissolved organic matter inflow drives temporal dynamics of the bacterial community of a subarctic estuary (northern Baltic Sea). *Environmental Microbiology* 23(8): 4200–4213. <https://doi.org/10.1111/1462-2920.15597>.
- Findlay, S., 2003. Bacterial Response to Variation in Dissolved Organic Matter. 363–379.
- Fonvielle, J. A., D. P. Giling, G. A. Singer, S. A. Berger, J. C. Nejtgaard, U. Mischke, A. Lychee Solheim, H.-P. Grossart & M. O. Gessner, 2016. Lake brownification and eutrophication effects on DOM dynamics. In: *EGU General Assembly Conference Abstracts*, p. EPSC2016–16929.
- Gonzalez, N., R. Anadon & L. Viesca, 2003. Carbon flux through the microbial community in a temperate sea during summer: role of bacterial metabolism. *Aquatic Microbial Ecology* 33(2): 117–126. <https://doi.org/10.3354/ame033117>.
- Goodman, K. J., M. A. Baker & W. A. Wurtsbaugh, 2011. Lakes as buffers of stream dissolved organic matter (DOM) variability: temporal patterns of DOM characteristics in mountain stream-lake systems. *Journal of Geophysical Research*. <https://doi.org/10.1029/2011jg001709>.
- Guenet, B., M. Danger, L. Abbadie & G. Lacroix, 2010. Priming effect: bridging the gap between terrestrial and aquatic ecology. *Ecology* 91(10): 2850–2861. <https://doi.org/10.1890/09-1968.1>.
- Hanson, P. C., D. L. Bade, S. R. Carpenter & T. K. Kratz, 2003. Lake metabolism: Relationships with dissolved organic carbon and phosphorus. *Limnology and Oceanography* 48(3): 1112–1119. <https://doi.org/10.4319/lno.2003.48.3.1112>.
- Holding, J. M., C. M. Duarte, A. Delgado-Huertas, K. Soetaert, J. E. Vonk, S. Agustí, P. Wassmann & J. J. Middelburg,

2017. Autochthonous and allochthonous contributions of organic carbon to microbial food webs in Svalbard fjords. *Limnology and Oceanography* 62(3): 1307–1323. <https://doi.org/10.1002/lno.10526>.
- Jansson, M., A.-K. Bergström, D. Lymer, K. Vrede & J. Karlsson, 2006. Bacterioplankton growth and nutrient use efficiencies under variable organic carbon and inorganic phosphorus ratios. *Microbial Ecology* 52(2): 358–364. <https://doi.org/10.1007/s00248-006-9013-4>.
- Jones, K., M. Berggren & J. Sjöstedt, 2023. Seasonal variation and importance of catchment area composition for transport of bioavailable carbon to the Baltic Sea. *Biogeochemistry*. <https://doi.org/10.1007/s10533-023-01079-y>.
- Kirchman, D. L., 1993. *Handbook of Methods in Aquatic Microbial Ecology*, vol Chapter 14, 1st edn. CRC Press, Boca Raton.
- Klante, C., M. Larson & K. M. Persson, 2021. Brownification in Lake Bolmen, Sweden, and its relationship to natural and human-induced changes. *Journal of Hydrology: Regional Studies* 36: 100863. <https://doi.org/10.1016/j.ejrh.2021.100863>.
- Kothawala, D. N., C. A. Stedmon, R. A. Muller, G. A. Weyhenmeyer, S. J. Kohler & L. J. Tranvik, 2014. Controls of dissolved organic matter quality: evidence from a large-scale boreal lake survey. *Global Change Biology* 20(4): 1101–1114. <https://doi.org/10.1111/gcb.12488>.
- Kritzborg, E. S., 2017. Centennial-long trends of lake browning show major effect of afforestation. *Limnology and Oceanography Letters* 2(4): 105–112. <https://doi.org/10.1002/lol2.10041>.
- Kritzborg, E. S., J. J. Cole, M. L. Pace, W. Graneli & D. L. Bade, 2004. Autochthonous versus allochthonous carbon sources of bacteria: Results from whole-lake C-13 addition experiments. *Limnology and Oceanography* 49(2): 588–596. <https://doi.org/10.4319/lo.2004.49.2.0588>.
- Kritzborg, E. S., W. Graneli, J. Björk, C. Brönmark, P. Hallgren, A. Nicolle, A. Persson & L.-A. Hansson, 2014. Warming and browning of lakes: consequences for pelagic carbon metabolism and sediment delivery. *Freshwater Biology* 59(2): 325–336. <https://doi.org/10.1111/fwb.12267>.
- Kritzborg, E. S., E. M. Hasselquist, M. Škerlep, S. Löfgren, O. Olsson, J. Stadmark, S. Valinia, L.-A. Hansson & H. Laudon, 2020. Browning of freshwaters: Consequences to ecosystem services, underlying drivers, and potential mitigation measures. *Ambio* 49(2): 375–390. <https://doi.org/10.1007/s13280-019-01227-5>.
- Lennon, J. T. & L. E. Pfaff, 2005. Source and supply of terrestrial organic matter affects aquatic microbial metabolism. *Aquatic Microbial Ecology* 39(2): 107–119. <https://doi.org/10.3354/ame039107>.
- Liess, A., C. Faithfull, B. Reichstein, O. Rowe, J. Guo, R. Pete, G. Thomsson, W. Uszko & S. N. Francoeur, 2015. Terrestrial runoff may reduce microbenthic net community productivity by increasing turbidity: a Mediterranean coastal lagoon mesocosm experiment. *Hydrobiologia* 753(1): 205–218. <https://doi.org/10.1007/s10750-015-2207-3>.
- Menzel, D. W. & N. Corwin, 1965. THE measurement of total phosphorus in seawater based on the liberation of organically bound fractions by persulfate oxidation I. *Limnology and Oceanography* 10(2): 280–282. <https://doi.org/10.4319/lo.1965.10.2.0280>.
- Meunier, C., M. Gundale, I. Sánchez & A. Liess, 2016. Impact of nitrogen deposition on forest and lake food webs in nitrogen limited environments. *Global Change Biology*. <https://doi.org/10.1111/gcb.12967>.
- Miranda, M. L., H. Osterholz, H. A. Giebel, P. Bruhnke, T. Dittmar & O. Zielinski, 2020. Impact of UV radiation on DOM transformation on molecular level using FT-ICR-MS and PARAFAC. *Spectrochimica Acta Part a: Molecular and Biomolecular Spectroscopy* 230: 118027. <https://doi.org/10.1016/j.saa.2020.118027>.
- Mormul, R. P., J. Ahlgren, M. K. Ekvall, L.-A. Hansson & C. Brönmark, 2012. Water brownification may increase the invasibility of a submerged non-native macrophyte. *Biological Invasions* 14(10): 2091–2099. <https://doi.org/10.1007/s10530-012-0216-y>.
- Moss, B., S. Kosten, M. Meerhoff, R. W. Battarbee, E. Jeppesen, N. Mazzeo, K. Havens, G. Lacerot, Z. Liu & L. De Meester, 2011. Allied attack: climate change and eutrophication. *Inland Waters* 1(2): 101–105.
- Murphy, K. R., C. A. Stedmon, D. Graeber & R. Bro, 2013. Fluorescence spectroscopy and multi-way techniques. PARAFAC. *Analytical Methods* 5(23): 6557–6566. <https://doi.org/10.1039/C3AY41160E>.
- Muscarella, M. E., X. M. Howey & J. T. Lennon, 2020. Trait-based approach to bacterial growth efficiency. *Environmental Microbiology* 22(8): 3494–3504. <https://doi.org/10.1111/1462-2920.15120>.
- Obernosterer, I. & R. Benner, 2004. Competition between biological and photochemical processes in the mineralization of dissolved organic carbon. *Limnology and Oceanography* 49(1): 117–124. <https://doi.org/10.4319/lo.2004.49.1.0117>.
- Pace, M. L. & Y. T. Prairie, 2005. 103Respiration in lakes. In del Giorgio, P. & P. Williams (eds), *Respiration in Aquatic Ecosystems*. Oxford University Press.
- Paczkowska, J., S. Brugel, O. Rowe, R. Lefebvre, A. Brutemark & A. Andersson, 2020. Response of Coastal Phytoplankton to High Inflows of Terrestrial Matter. *Frontiers in Marine Science*. <https://doi.org/10.3389/fmars.2020.00080>.
- Pennanen, V. K., Pirkko; Mannio, Jaakko, 1986. Comparative study on the estimation of humic matter in natural waters. Vesihallitus. National Board of Waters.
- Pérez, M. T. & R. Sommaruga, 2006. Differential effect of algal- and soil-derived dissolved organic matter on alpine lake bacterial community composition and activity. *Limnology and Oceanography* 51(6): 2527–2537. <https://doi.org/10.4319/lo.2006.51.6.2527>.
- Sand-Jensen, K. & P. A. Staehr, 2009. Net heterotrophy in small Danish Lakes: a widespread feature over gradients in trophic status and land cover. *Ecosystems* 12(2): 336–348. <https://doi.org/10.1007/s10021-008-9226-0>.
- Sawyer, C., P. McCarty & G. Parkin, 1978. *Biochemical oxygen demand. Chemistry for environmental engineering*, 416–432, McGraw-Hill, New York.
- Smith, D. C. & F. Azam, 1992. A simple, economical method for measuring bacterial protein synthesis rates in seawater using 3H-leucine. In, 1992.

- Smith, E. M. & Y. T. Prairie, 2004. Bacterial metabolism and growth efficiency in lakes: the importance of phosphorus availability. *Limnology and Oceanography* 49(1): 137–147. <https://doi.org/10.4319/lo.2004.49.1.0137>.
- Soares, A. R. A., E. S. Kritzbeg, I. Custelcean & M. Berggren, 2018. Bacterioplankton responses to increased organic carbon and nutrient loading in a boreal estuary—separate and interactive effects on growth and respiration. *Microbial Ecology* 76(1): 144–155. <https://doi.org/10.1007/s00248-017-1115-7>.
- Soares, A. R. A., J.-F. Lapierre, B. P. Selvam, G. Lindström & M. Berggren, 2019. Controls on Dissolved Organic Carbon Bioreactivity in River Systems. *Scientific Reports*. <https://doi.org/10.1038/s41598-019-50552-y>.
- Sondergaard, M. & M. Middelboe, 1995. A cross-system analysis of labile dissolved organic-carbon. *Marine Ecology Progress Series* 118(1–3): 283–294. <https://doi.org/10.3354/meps118283>.
- Spilling, K., E. Asmala, N. Haavisto, L. Haraguchi, K. Kraft, A.-M. Lehto, A. M. Lewandowska, J. Norkko, J. Piiparinen, J. Seppälä, M. Vanharanta, A. Vehmaa, P. Ylöstalo & T. Tamminen, 2022. Brownification affects phytoplankton community composition but not primary productivity in eutrophic coastal waters: a mesocosm experiment in the Baltic Sea. *Science of the Total Environment* 841: 156510. <https://doi.org/10.1016/j.scitotenv.2022.156510>.
- Taipale, S. J., K. Vuorio, U. Strandberg, K. K. Kahilainen, M. Järvinen, M. Hiltunen, E. Peltomaa & P. Kankaala, 2016. Lake eutrophication and brownification downgrade availability and transfer of essential fatty acids for human consumption. *Environment International* 96: 156–166. <https://doi.org/10.1016/j.envint.2016.08.018>.
- Taipale, S. J., K. K. Kahilainen, G. W. Holtgrieve & E. T. Peltomaa, 2018. Simulated eutrophication and browning alters zooplankton nutritional quality and determines juvenile fish growth and survival. *Ecology and Evolution* 8(5): 2671–2687. <https://doi.org/10.1002/ece3.3832>.
- Temnerud, J., J. K. Hytteborn, M. N. Futter & S. J. Köhler, 2014. Evaluating common drivers for color, iron and organic carbon in Swedish watercourses. *Ambio* 43(S1): 30–44. <https://doi.org/10.1007/s13280-014-0560-5>.
- Thompson, S. K. & J. B. Cotner, 2020. P-limitation drives changes in DOM production by aquatic bacteria. *Aquatic Microbial Ecology* 85: 35–46.
- Tranvik, L. J., J. A. Downing, J. B. Cotner, S. A. Loiselle, R. G. Striegl, T. J. Ballatore, P. Dillon, K. Finlay, K. Fortino, L. B. Knoll, P. L. Kortelainen, T. Kutser, S. Larsen, I. Laurion, D. M. Lee, S. L. McCallister, D. M. McKnight, J. M. Melack, E. Overholt, J. A. Porter, Y. Prairie, W. H. Renwick, F. Roland, B. S. Sherman, D. W. Schindler, S. Sobek, A. Tremblay, M. J. Vanni, A. M. Verschoor, E. Von Wachenfeldt & G. A. Weyhenmeyer, 2009. Lakes and reservoirs as regulators of carbon cycling and climate. *Limnology and Oceanography* 54(6): 2298–2314. [https://doi.org/10.4319/lo.2009.54.6\\_part\\_2.2298](https://doi.org/10.4319/lo.2009.54.6_part_2.2298).
- Traving, S. J., O. Rowe, N. M. Jakobsen, H. Sorensen, J. Dinasquet, C. A. Stedmon, A. Andersson & L. Riemann, 2017. The Effect of Increased Loads of Dissolved Organic Matter on Estuarine Microbial Community Composition and Function. *Frontiers in Microbiology*. <https://doi.org/10.3389/fmicb.2017.00351>.
- Tumdedo, M. E., 2010. Dynamics of Humic Substance in Bolmen Lake, Sweden. In: University, L. (ed) Master Thesis.
- Unanue, M., B. Ayo, M. Agis, D. Slezak, G. J. Herndl & J. Iriberry, 1999. Ectoenzymatic activity and uptake of monomers in marine bacterioplankton described by a biphasic kinetic model. *Microbial Ecology* 37(1): 36–48.
- Urrutia-Cordero, P., O. Langvall, P. Blomkvist, D. G. Angeler, S. Bertilsson, W. Colom Montero, P. Eklöv, N. Aagaard Jakobsen, L. Klemetsson, H. Laudon, B. Liljebladh, M. Lundgren, L. Parkefelt, E. Kelpsiene, D. Pierson, J. Rankinen, M. Striebel, L. J. Tranvik, P. Weslien, H. Hillebrand & S. Langenheder, 2021. SITES AquaNet: An open infrastructure for mesocosm experiments with high frequency sensor monitoring across lakes. *Limnology and Oceanography: Methods* 19(6): 385–400. <https://doi.org/10.1002/lom3.10432>.
- Wilhelm, F. M., 2009. Pollution of Aquatic Ecosystems I. 110–119.
- Woolway, R. I., B. M. Kraemer, J. D. Lenters, C. J. Merchant, C. M. O'Reilly & S. Sharma, 2020. Global lake responses to climate change. *Nature Reviews Earth & Environment* 1(8): 388–403. <https://doi.org/10.1038/s43017-020-0067-5>.
- Wünsch, U. J. & K. Murphy, 2021. A simple method to isolate fluorescence spectra from small dissolved organic matter datasets. *Water Research* 190: 116730. <https://doi.org/10.1016/j.watres.2020.116730>.
- Zhao, L., S. Brugel, K. P. Ramasamy & A. Andersson, 2023. Bacterial community responses to planktonic and terrestrial substrates in coastal northern Baltic Sea. *Frontiers in Marine Science*. <https://doi.org/10.3389/fmars.2023.1130855>.
- Zweifel, U., B. Norrman & Å. Hagström, 1993. Consumption of dissolved organic carbon by marine bacteria and demand for inorganic nutrients. *Marine Ecology Progress Series* 101: 23–32. <https://doi.org/10.3354/meps101023>.

**Publisher's Note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.